

Thirty Years of Service Failure and Recovery Research: Thematic Development and Future Research Opportunities From a Social Network Perspective

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Abstract

Service failure and recovery (SFR) is a well-established area of research that has made considerable progress over the past 30 years. In this study, we used a combination of text mining, co-word analysis, and social network analysis (SNA) to explore the relationships among keywords in SFR research. We analyzed a dataset of 533 SFR articles published between 1990 and 2020, extracting the most frequently used keywords using text-mining techniques. These keywords were then subjected to co-word analysis and SNA to understand the development of themes and topics in SFR research. By examining changes in network centrality measures, we gained insights into the evolution of research in this field. Furthermore, by identifying gaps or disconnections in the keyword networks, we identified future research opportunities related to the impact of service recovery strategies on customer reactions, employee reactions, and firm outcomes, as well as the relationship between customer and employee responses.

Keywords

service failure, service recovery, keyword networks, co-word analysis, social network analysis

Introduction

Service failure and recovery (SFR) has garnered considerable attention from scholars across various disciplines for the past three decades. Service failure refers to situations wherein organizations fail to meet consumer needs and expectations, while service recovery involves the organization's response to such failures (Tax and Brown 1998; Van Vaerenbergh et al. 2019). In recent years, service failures have become increasingly prevalent, as evidenced by rising consumer complaints. For instance, according to the Consumer Federation of America, consumer agencies reported 208,000 complaints in 2021, which escalated to 600,000 in 2022 (Consumer Federation of America 2021, 2022). Service failures were prominently observed in categories such as auto sales and repair, home improvement repairs, and financial services (Consumer Federation of America 2022). The repercussions of service failures are significant, leading to diminished customer satisfaction and loyalty (Hess et al., 2003). However, effective service recovery can yield positive outcomes. When implemented properly, service recovery mitigates customers' negative reactions to service failures and enhances their satisfaction (Smith and Bolton 1998). Conversely, mishandled service recovery efforts can result in customer dissatisfaction and attrition (McCollough, Berry, and Yadav 2000).

Consistent with its practical significance, SFR has become an extensively researched topic. Multiple streams of research have emerged within the realm of SFR, exploring areas such as customer satisfaction (De Matos et al., 2007; Orsingher et al., 2010), customer participation (Van Vaerenbergh, Hazée, and Costers 2018), customer emotions (Valentini, Orsingher, and Polyakova 2020), service recovery responses (Roschk and Gelbrich 2014), service failure attributions (Van Vaerenbergh et al., 2014), and online service recovery (Manu C and Sreejesh S 2020). Two recent review articles using bibliometric methods have been instrumental to an improved understanding of knowledge foundational to the SFR literature. One of these review articles, Fouroudi et al. (2020), conducted co-citation

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analysis on 416 SFR articles, identifying nine prominent research themes including customer satisfaction, consumer emotional responses, consumer behavioral responses, and justice theory. Similarly, [Arora and Chakraborty \(2021\)](#) used co-citation analysis to uncover 12 knowledge clusters in consumer complaining behavior literature, ranging from consumer cognitive evaluation to emotional reactions and complaint management. The current research serves as a valuable addition to existing bibliometric reviews on SFR and contributes to our understanding of SFR in several important ways.

First, co-word analysis offers a valuable tool for examining the intellectual structure of SFR research at a granular level. While co-citation analysis allows us to identify overarching research themes by analyzing article-level citations and co-citations, co-word analysis focuses on the specific keywords used in each article. This approach enables us to examine the actual content of publications, providing insights into the concepts and variables that have been explored. In their study, [Donthu et al. \(2021\)](#) emphasize the importance of utilizing co-word analysis alongside co-citation analysis, as it offers a complementary perspective by focusing on the actual content of the research.

By employing co-word analysis, we gain a better understanding of the specific concepts and variables that have garnered attention in SFR research. For example, by examining the top keywords and their relative importance, we observe relatively strong interest among scholars in exploring consumer emotional and behavioral responses, such as anger, revenge, and forgiveness, as well as specific recovery strategies, including explanation, tangible and intangible compensation, apology, and customer participation. Interestingly, our analysis revealed a relative lack of emphasis on employee-related variables in SFR research, indicating a potential avenue for future exploration. To the best of our knowledge, this study is the first to leverage co-word analysis to provide a nuanced understanding of the SFR literature. By delving into specific keywords and their interrelationships, we contribute to a clearer view of the landscape in this research domain.

Second, our study takes a longitudinal approach to examine the SFR literature. Over the past three decades, SFR research has established itself as a prominent area within the service literature. However, it is important to recognize that research interests and foci evolve over time. While previous co-citation analyses conducted by [Arora and Chakraborty 2021](#); [Fouroudi et al., 2020](#) provided valuable insights into well-explored topics, the change in research emphases within SFR literature across time has not been investigated.

To address this limitation, we divided the literature into three distinct time periods and analyzed the changes in the relative importance of keywords across these periods. This approach enabled us to track the evolution of SFR literature and identify emerging research topics. Our findings, based on SNA, reveal noteworthy shifts in research interests since 2015. For example, we observed significant increases in network centrality measures for concepts (keywords) such as negative word-of-mouth, online complaints, perceived justice, and customer loyalty. These upward trends indicate growing scholarly attention paid to these (and associated) phenomena in recent years. By taking a longitudinal perspective, our study not only sheds light on the

current state of SFR literature but also provides unique insight into the dynamic nature of research foci within the field.

Third, co-word analysis offers a unique advantage over previous reviews in guiding future research directions in SFR. Citation and co-citation analyses, while valuable, tend to be retrospective in nature, often excluding recent or niche publications that have yet to accumulate a substantial number of citations ([Donthu et al., 2021](#)). In contrast, co-word analysis, when combined with SNA, allows us to examine both the existing associations (network connections) and the unexplored areas (network disconnections or “structural holes”) in SFR research.

This approach proves particularly suitable for providing future directions and generating specific research questions ([Donthu et al., 2021](#)). By identifying the network connections and disconnections among keywords, we gain valuable insights into the potential avenues for further investigation. For instance, our SNA reveals a gap in the literature concerning the relationship between apology and the work engagement of service employees. As a result, we propose that future research explores the effects of offering apologies on work engagement (and the potential mediators and moderators influencing this relationship). In total, our analysis has generated more than 30 specific research questions, each addressing the structural holes identified in the keyword network from the years 2016 to 2020. By pinpointing these research gaps and explicitly encouraging future researchers to delve into these unexplored areas, we contribute to the advancement of SFR scholarship.

Fourth, in our pursuit of a comprehensive review of SFR research, we carried out three supplementary analyses to fill in the gaps. One of our supplementary analyses focused on examining the research methods used in SFR research. Through this analysis, we observed an increasing trend in the utilization of multiple research methods within the SFR field. The other two supplementary analyses centered around identifying the trending topics in recent SFR research. Notably, we observed a surge of interest in AI-enabled self-services as a prominent theme in the field. Combining the insights from (1) existing bibliometric reviews with co-citation analysis, (2) our main analysis involving co-word analysis and SNA, and (3) the three supplementary analyses, we gain a more comprehensive understanding of SFR literature and its future development.

The paper is structured into several sections. First, we describe the data collection and analytical procedures for our co-word analysis. Subsequently, we present the insights derived from analyzing changes in keywords. Following that, we propose future research opportunities that highlight understudied connections in SFR literature. Additionally, we conduct three supplementary analyses to further enhance understanding of the field. Finally, we conclude the paper and discuss its limitations.

Research Method

Data Collection

We followed a systematic approach to identify relevant literature for our bibliometric analysis. Firstly, we compiled a list of

46 journals from various fields, including marketing, management, hospitality, and psychology, based on the article list provided by Fouroudi et al. (2020). These journals were likely to feature research related to SFR. Secondly, we conducted comprehensive searches across multiple databases such as ABI, EBSCO, Web of Science, and journal websites, using combinations of keywords such as “service failure” and “service recovery.” This ensured that we covered all potential SFR articles published in the selected journals. We carefully removed any duplicates that might have appeared in multiple databases. This process yielded 1,200 articles. Thirdly, we screened each of the 1,200 articles to ensure that SFR was the central focus in each publication. Following established practices, we included review articles and meta-analysis studies while excluding book chapters, book reviews, and editorials from our dataset. After this screening process, we ended up with a final collection of 533 articles published between 1990 and 2020.

Among the selected journals, the top three sources that contributed the most articles to our dataset were the *Journal of Services Marketing* (59 articles), *Journal of Service Research* (54 articles), and *Journal of Service Management* (48 articles). Table 1 provides an overview of the top 10 journals in our database. Figure 1 displays the distribution of SFR articles across the 30-year period. The upward trend depicted in Figure 1 reflects the increasing

attention and significance given to the topic of SFR over the past three decades. To capture the evolution of SFR research, we divided the sample into three distinct periods: 1990-2009 (156 articles), 2010-2015 (166 articles), and 2016-2020 (211 articles). These periods were selected based on the criteria of having enough studies published in each period and maintaining a balanced distribution of articles among the periods.

Keyword Extraction and Standardization

Given that many SFR articles did not have author-provided keywords, we employed a text mining approach with Python to extract keywords from the 533 SFR articles. This technique allowed us to identify the most common words and phrases in each article, resulting in 5161 keywords, averaging around nine keywords per article. After the keyword extraction, we proceeded with keyword standardization to account for variations in expressions that convey the same idea or concept. For instance, we consolidated related keywords such as “complaining behavior,” “complaints,” “complaint voicing,” and “dissatisfied complaining” under the term “complaining behavior.” This standardization process ensured consistency and accuracy in our analysis. As a result, we identified the top 25 most frequent keywords for 1990-2009, 31 for 2010-2015, and 33 for

Table 1. Journals With the Most SFR Articles From 1990 to 2020.

Ranking	Journal Name	Quantity
1	Journal of Services Marketing	59
2	Journal of Service Research	54
3	Journal of Service Management	48
4	International Journal of Contemporary Hospitality Management	41
5	Journal of Business Research	41
6	European Journal of Marketing	35
7	Service Industries Journal	30
8	Journal of the Academy of Marketing Science	26
9	Journal of Retailing and Consumer Services	22
10	Journal of Hospitality Marketing and Management	20

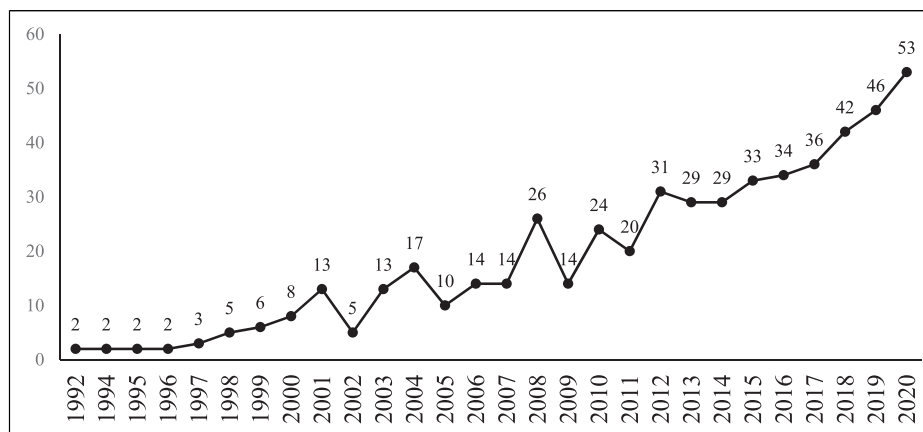


Figure 1. The number of SFR research articles by year.

2016–2020. These keywords represent the prevalent themes and concepts explored in SFR research during each respective time frame.

Keyword Co-occurrence Matrices and SNA

Co-word analysis involves constructing a co-occurrence matrix that quantifies how frequently pairs of keywords appear together in the same article. This matrix reflects the strength of connections between keywords. For example, we observed in the 2016–2020 matrix that “customer satisfaction” had a stronger connection with “perceived justice” ($n = 11$) than with “leadership” ($n = 0$).

Taking a social network perspective, we analyzed the SFR literature as an evolving system of interconnected entities. Using UCINET, we assessed keyword centralities and visualized the relationships between keywords. Figure 2 illustrates the keyword network structure in the 2016–2020 period, while Figure 1WA and Figure 2WA in the web appendix show the structures in the other two earlier periods. The size of each node in the visualization represents the closeness centrality of the respective keyword, while the thickness of the lines indicates the strength of the connections.

Network density is an overall network-level characteristic measuring “the probability that a tie exists between any pair of randomly chosen nodes” (Borgatti et al. 2013, p. 150). Our UCINET results revealed densities of 1.010 for the 1990–2009 period, 0.819 for the 2010–2015 period, and 0.765 for the 2016–2020 period. This evolution is understandable as the field matures. In any research field, we can consider the

relationships among core variables, such as customer satisfaction and perceived justice in SFR, as the “stem.” As this stem matures, new “branches” have started to emerge and grow. The relatively low network density of keywords in recent SFR literature suggests that existing research has not thoroughly explored the connections among these new branches, presenting valuable opportunities for future investigations.

Closeness centrality is a measure of the relative importance of each keyword in a co-word analysis (Ronda-Pupo and Guerras-Martin 2012). Our results show that the keyword closeness centrality values ranged from 0.381 to 0.889 in the 1990–2009 period, from 0.462 to 0.882 in the 2010–2015 period, and from 0.464 to 0.800 in the 2016–2020 period (refer to Table 2). Following recommendations from Ronda-Pupo and Guerras-Martin (2012), we categorized these keywords into three groups: core, semi-periphery, and periphery, based on their centrality values.

CO-Word Analysis Findings: Thematic Development in SFR

Table 2 provides keyword network centrality measures that reveal numerous changes over time, including the emergence, disappearance, and shifts of keywords. Considering the large number of identified keywords, we will not delve into the details of each individual change presented in Table 2. Instead, we will highlight a selection of intriguing and insightful findings.

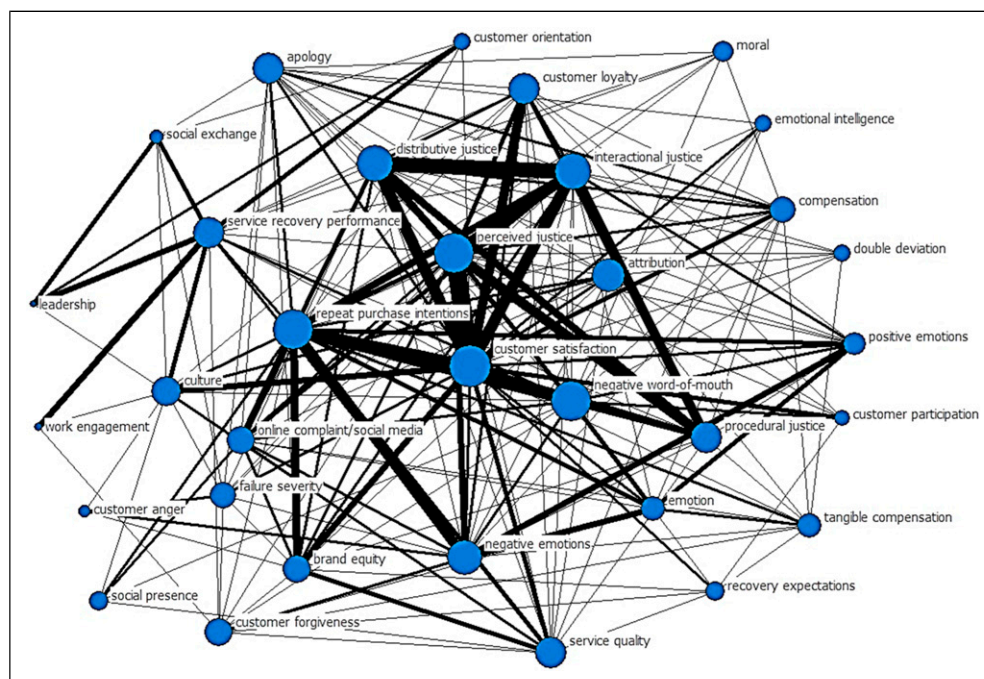


Figure 2. Social network for SFR literature in the 2016–2020 period. Notes: Node size is based on closeness centrality; line thickness is based on tie strength.

Table 2. Most Frequent Keywords in SFR Literature.

Category	1990-2009	Closeness Centrality	2010-2015	Closeness Centrality(Δ)	2016-2020	Closeness Centrality(Δ)
Core	Customer satisfaction	0.889	Customer satisfaction	0.882 (-0.007)	Customer satisfaction	0.800 (-0.082)
	Service quality	0.727	Negative emotions	0.75 (+0.135)	Perceived justice/fairness	0.78 (+0.098)
					Repurchase intentions	0.78 (+0.082)
					Negative word-of-mouth	0.762 (+0.064)
					Distributive justice/fairness	0.711 (+0.034)
					Interactional justice/fairness	0.711 (+0.029)
					Negative emotions	0.696 (-0.054)
Semi-periphery	Distributive justice/fairness	0.686	Failure severity	0.698 (+0.113)	Attribution	0.681 (+0.069)
	Interactional justice/fairness	0.686	Negative word-of-mouth	0.698 (+0.066)	Apology*	0.667
	Customer expectations	0.667	Repurchase intentions	0.698 (+0.031)	Recovery performance	0.667 (+0.067)
	Repurchase intentions	0.667	Culture	0.682 (+0.097)	Customer loyalty	0.653 (+0.053)
	Negative word-of-mouth	0.632	Interactional justice/fairness	0.682 (-0.004)	Procedural justice/fairness	0.653 (-0.029)
	Negative emotions	0.615	Perceived justice/fairness	0.682 (+0.067)	Service quality	0.653 (+0.015)
	Perceived justice/fairness	0.615	Procedural justice/fairness	0.682 (+0.097)	Culture	0.640 (-0.042)
	Culture	0.585	Distributive justice/fairness	0.667 (-0.019)	Brand equity	0.627 (+0.091)
	Compensation	0.585	Switching costs	0.652 (+0.119)	Online complaint/social media	0.627 (+0.091)
	Failure severity	0.585	Relationship management*	0.625	Customer forgiveness*	0.615
	Procedural justice/fairness	0.585	Service quality	0.638 (-0.102)	Compensation	0.604 (+0.019)
	Critical incident	0.571	Attribution*	0.612	Failure severity	0.604 (-0.094)
	Recovery performance	0.571			Tangible compensation	0.593 (+0.016)
	Organizational commitment	0.558			Emotion	0.582 (-0.006)
Periphery	Customer loyalty	0.545	Attitude*	0.6	Positive emotions*	0.571
	Relationship quality	0.533	Customer loyalty	0.6 (+0.055)	Moral*	0.552
	Service guarantee	0.533	Recovery performance	0.6 (+0.029)	Recovery expectations*	0.542
	Switching costs	0.533	Trust/commitment	0.6 (+0.129)	Social presence*	0.542
	Control	0.522	Emotion*	0.588	Customer orientation*	0.533
	Staff empowerment	0.511	Tangible compensation*	0.577	Double deviation*	0.525
	Job satisfaction	0.471	Explanation*	0.556	Emotional intelligence	0.525 (-0.020)
	Trust/commitment	0.471	Customer revenge*	0.545	Customer participation*	0.516
	Role stress	0.381	Emotional intelligence*	0.545	Social exchange*	0.508
			Intangible compensation*	0.536	Customer anger	0.492 (-0.034)
			Brand equity*	0.536	Work engagement*	0.471

(continued)

Table 2. (continued)

Category	1990-2009	Closeness Centrality	2010-2015	Closeness Centrality(Δ)	2016-2020	Closeness Centrality(Δ)
			Online complaint/ social media*	0.536	Leadership*	0.464
			Customer anger*	0.526		
			Behavioral intentions*	0.508		
			Critical incident	0.508 (–0.063)		
			Service provision*	0.492		
			Job satisfaction	0.462 (–0.009)		

Notes: Numbers in parentheses are the changes in closeness centrality measures. *Keywords in a period list that were not in the prior period.

The Rising Prevalence of Negative Word-Of-mouth and Online Complaints

The advent of social media platforms and online channels has empowered consumers to propagate negative word-of-mouth experiences, resulting in heightened significance of this phenomenon. The closeness centrality of the “negative word-of-mouth” keyword has exhibited a steady progression from 0.632 during the 1990-2009 period to 0.698 in the 2010-2015 period, eventually growing to 0.762 during the 2016-2020 period. Consequently, negative word-of-mouth has emerged as a topic “core” to the SFR literature during the latter period. Furthermore, the “online complaint” keyword initially surfaced on the keyword network “periphery” during 2010-2015 period but has since grown to the network’s “semi-periphery” during the 2016–2020 period, with closeness centrality increasing from 0.536 to 0.627.

A More Intricate Comprehension of Consumer Emotional and Behavioral Responses

Table 2 provides evidence of an expanding knowledge base regarding specific categories of consumer emotional and behavioral responses within the SFR context. For instance, compared to the 1990-2009 period, the emergence of several new keywords associated with consumer emotional and behavioral responses, such as anger and revenge, became apparent in the 2010-2015 period. Similarly, the 2016-2020 period witnessed the emergence of additional relevant keywords, including forgiveness and positive emotions.

An Enhanced Understanding of Various Types of Recovery Strategies

In the 1990-2009 period, compensation stood as the sole prominent keyword closely linked to recovery strategies. However, the emphasis on recovery strategies expanded in latter periods, as evidenced by the emergence of keywords such as “explanation,” “tangible compensation,” and “intangible compensation” during the 2010-2015 period, and subsequently, “apology” and “customer participation” during the 2016-2020 period.

Insufficient Attention to Employee-related Variables

Despite the enriched understanding of customer reactions and recovery strategies, our analysis suggests that insufficient scholarly attention has been dedicated to employee-related variables. During the 1990-2009 period, three keywords associated with service employees, namely, “job satisfaction,” “organizational commitment,” and “role stress,” stood among the literature’s most prominent. However, both organizational commitment and role stress fell off the lists of most frequent keywords, leaving job satisfaction as the sole variable clearly linked to employees during the 2010-2015 period. Furthermore, job satisfaction itself ceased to be a focal keyword during the 2016-2020 period, with “work engagement” seemingly taking its place as the sole employee-related variable.

In addition to the aforementioned findings, several other noteworthy observations emerge. For example, justice-related keywords, including perceived justice/fairness, distributive justice/fairness, and interactional justice/fairness, have transitioned from the network’s periphery to the network’s core, with their centralities increasing between the 2010-2015 and 2016-2020 periods. Moreover, scholars have exhibited a heightened focus on loyalty as an outcome of SFR, as indicated by the increased centrality of the “repurchase intentions” and “customer loyalty” keywords during the 2016-2020 period. Relatedly, it is worth noting that “customer satisfaction” has remained as the keyword most central (or most “core”) to the SFR literature across the time periods examined.

Future Research Opportunities

Utilizing SNA results from the 2016–2020 period, we developed a model to identify potential avenues for future research (refer to Figure 3). To create a parsimonious conceptual model, we organized similar keywords into broader categories. For instance, keywords like apology, compensation, and customer participation were grouped under the broader category of “service recovery strategy.” Additionally, other keywords contributed to three response variable categories: customer reactions, employee reactions, and firm outcomes. Figure 3 utilizes recent SFR literature to illustrate the mechanisms through which a firm’s service recovery strategies influence its

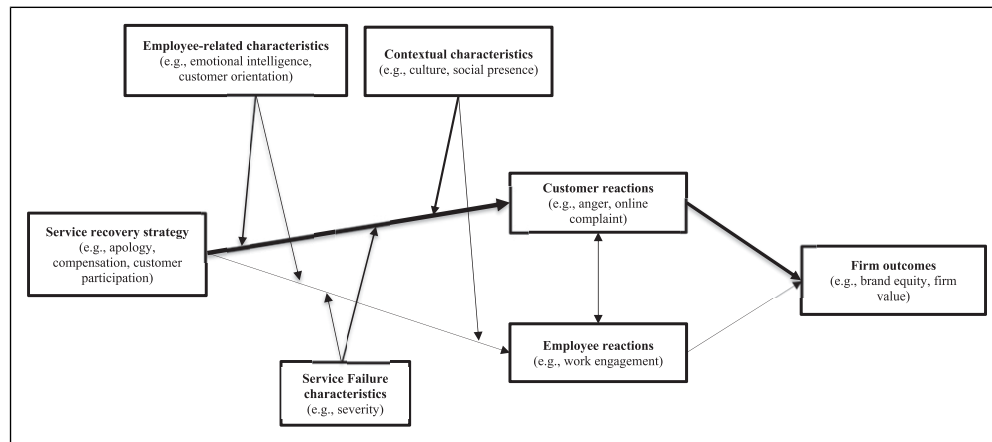


Figure 3. The model for future directions based on the 2016-2020 SNA results. Note. The direction of the arrow is based on the extant literature. The thickness of the arrow represents the strength of the linkages between/among connected groups, which is calculated based on the keyword co-occurrence matrix. More information about the calculation is available upon request. The thicker an arrow, the stronger the linkage is (more literature has studied the relationship).

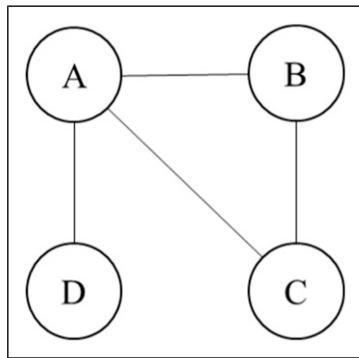


Figure 4. A simplified keyword network.

own performance. Figure 3 also highlights that the impact of service recovery strategies is influenced by three sets of keywords: employee characteristics, contextual characteristics, and service failure characteristics. The thickness of an arrow in Figure 3 indicates the extent to which relationships among categories of variables have been explored together within the extant SFR literature; thinner arrows prompt us to consider underexplored ideas that present valuable research opportunities. For instance, there has been more emphasis on studying the effects of service recovery strategies on customer reactions compared to employee reactions.

In SNA, structural holes occur when there is no direct connection between two nodes (Burt 1995). In the context of a keyword network, a structural hole (operationalized here as a “0” in a keyword co-occurrence matrix) indicates an underexplored link between two keywords. For instance, in Figure 2, a structural hole is observed between “leadership” and “double deviation,” suggesting that SFR research has yet to fully investigate the relationship between these variables. Consequently, within keyword networks, structural holes can represent valuable literature gaps and suggest future research questions (Schrock et al. 2018). As shown in a simplified social

network model (Figure 4) where A, B, C, and D indicate keywords, B and C are not directly connected to D, suggesting the existence of a structural hole. We may derive at least three potential research opportunities here. Future research may explore (1) the understudied direct impacts of D on C and D on B; (2) D as a potential mediator between B and C; and (3) D as a potential moderator on the relationship between B and C.

By combining the insights from Figure 3 and an analysis of the structural holes in the 2016-2020 keyword network, we can derive meaningful directions and opportunities for future research. Some of the future directions described in the section below are summarized in Table 3.

Research Opportunities Related to the Impact of Service Recovery Strategy on Customer Reactions

Despite the predominant focus of prior studies on customer reactions to service recovery strategies, our co-word analysis reveals several promising avenues for future research.

Apology or Customer Participation → Customer Forgiveness? Our co-word analysis reveals the emergence of certain keywords related to service recovery strategies and customer reactions. Notably, apology and customer participation have gained attention as important service recovery strategies, while customer forgiveness and positive emotions have emerged as significant customer reactions. However, our analysis shows a disconnection between “customer forgiveness” and both “customer participation” and “apology” in Figure 2. This indicates a gap in the literature regarding the effectiveness of these strategies in facilitating customer forgiveness. Understanding the factors that influence customer forgiveness following service failures is of practical importance, as it can help firms mend relationships with their customers and reduce negative emotions associated with service failures (Tsarenko, Strizhakova, and Otnes 2019). Therefore, future studies exploring the relative efficacy of these

Table 3. Summary of Future Research Opportunities and Research Questions.**Research Stream 1: The impact of service recovery strategy on consumer reactions**

Apology → customer forgiveness	- Which types of apologies (e.g., forced vs. sincere, online vs. in-person) are more likely to elicit customer forgiveness? - What psychological mechanisms mediate the effect of apologies on customer forgiveness? - What factors moderate the effect of apologies on customer forgiveness? Are customer orientation, employee emotional intelligence, and failure severity possible moderators?
Apology → customer anger	- Can apologies effectively reduce customer anger? Which types of apologies are more effective in mitigating customer anger? Under what conditions might apologies actually increase customer anger? - In combination with apologies, which service recovery strategies should be employed to effectively reduce customer anger? - What psychological mechanisms mediate the effect of apologies on customer anger? - What factors moderate the effect of apologies on customer anger? Are customer orientation, employee emotional intelligence, and failure severity possible moderators?
Customer participation → customer forgiveness	- Does customer participation have a positive or negative impact on customer forgiveness? - Are perceived fairness, transparency, customer investment, and customer expectation potential mediators on this relationship? - Are customer orientation, employee emotional intelligence, and failure severity possible moderators on this relationship?

Research Stream 2: The impact of service recovery strategy on service employees

Exploring various employee reactions	- How do service failures and recovery strategies influence service employee brand identification and brand-building behaviors? - How do service recovery strategies and service employee traits like grit and resilience, and job meaningfulness interact with each other? - How do service recovery strategies and service employee job meaningfulness influence each other? - How do service recovery strategies and service employee ethics influence each other?
Apology → work engagement	- Does offering apologies have a positive or negative impact on employee work engagement? - What cognitive or affective mechanisms explain the (positive or negative) effects of employees offering apologies on their work engagement? - How do different types of apologies affect work engagement? - How can apologies be effectively utilized to generate positive customer reactions without imposing significant emotional labor on service employees?

Research Stream 3: The relationship between consumer reactions and employee reactions

Customer anger ↔ work engagement	- Does customer anger strengthen or undermine employee work engagement? - Can perceived fairness or employee motivation mediate the impact of customer anger on employee work engagement? - Can a high level of employee work engagement mitigate customer anger?
Exploring contingencies	- Does the relationship between customer reactions and employee reactions depend on leadership styles (e.g., directive vs. participative)? - Is the relationship between customer reactions and employee reactions stronger or weaker in an online service setting compared to an offline setting?

Research Stream 4: The impact of service recovery strategy on firm outcomes

Double deviation → customer reactions → employee reactions → brand equity	- Does double deviation have a negative impact on a company's brand equity? - Do specific customer reactions (e.g., frustration, revenge) mediate the impact of double deviation on brand equity? - Do specific employee reactions (e.g., frustration, burnout) mediate the impact of customer reactions on brand equity?
When SFR meets Wall Street	- How does SFR impact firm stock prices and firm value? - Is the impact of SFR on firm value amplified by social media?

strategies in inducing customer forgiveness would make a valuable contribution to the field.

Apology → Customer Anger? Another structural hole is observed between the “apology” and “customer anger”

keywords. Apologies can demonstrate respect and politeness towards customers who have experienced a service failure. However, it is crucial to gain a deeper understanding of the relationship between apology and customer anger to ensure that apologies are utilized in a manner that does not

exacerbate the negative situation. For example, future research can examine various types of apologies (e.g., forced vs. sincere, online vs. in-person). Customers might only grow angrier when they realize that frontline employees have been forced to apologize, when they perceive the apology to be insincere, or when they receive an automated apology transmitted through digital platforms.

Customer Participation → Customer Forgiveness? The presence of a structural hole between customer participation and customer forgiveness indicates still-limited understanding within this area, leaving us with insufficient knowledge about how customer participation in service impacts customer forgiveness after a service failure. On the one hand, when customers actively engage in the service recovery process, they may perceive a fairer resolution (Van Vaerenbergh, Hazée, and Costers 2018) and value the transparency (Balaji et al. 2018), which can foster forgiveness. However, as customers invest more time and effort in the recovery process, their expectations may also become elevated (Bagherzadeh et al., 2020). If these heightened expectations are not somehow met, customer participation could lead to feelings of anger instead. Therefore, understanding the (conditional) interplay between customer participation and customer anger/forgiveness remains an intriguing avenue for further research in the SFR literature.

Exploring the Contingencies. In order to uncover potential factors that may influence the relationship between service recovery strategies and customer reactions, future research can leverage Figure 2 as a guide. The figure highlights certain variables, such as customer orientation, failure severity, and emotional intelligence, that are not strongly connected with the identified recovery strategies and customer reactions (e.g., apology, customer participation, anger, and forgiveness). As a result, these variables present opportunities for further investigation as potential moderators of these relationships. For instance, future studies could explore whether the effectiveness of apology and customer participation in fostering customer forgiveness is significantly influenced by the level of customer orientation exhibited by the service company. It is plausible that customer orientation (which could be measured at multiple levels) enhances these recovery strategies' impact on customer forgiveness.

Similarly, emotional intelligence, as a personal attribute of service employees, could also play a role in moderating the relationship between recovery strategies and customer reactions. It would be interesting to examine whether service employees with higher (lower) levels of emotional intelligence are more (less) effective in using apologies and customer participation to engender customer forgiveness. By delving into these potential moderators, researchers can shed important light on the contextual conditions and individual characteristics that may enhance or diminish the effects of recovery strategies on customer reactions.

Research Opportunities Related to the Impact of Service Recovery Strategy on Service Employees

As indicated by the co-word analysis findings (finding #4), there has been a noticeable lack of research focusing on employee reactions to SFR. Over the span of 30 years, only a limited number of variables related to employees (job satisfaction, organizational commitment, role stress, work engagement) have been thoroughly investigated in SFR research. Furthermore, Figure 3, which specifically examines the 2016-2020 SFR literature, shows that the impact of service recovery strategies on employee reactions has received considerably less attention compared to its impact on customer reactions. Given the vital importance of employees in effectively handling service failures and implementing successful recovery strategies, it is crucial to deepen our understanding of their reactions and experiences.

More Research on Various Employee-Related Variables. As we move forward, there is an opportunity for future research to explore service employees' emotional and behavioral responses within the SFR context, beyond the aforementioned employee-related variables extensively explored in SFR research. For instance, studies investigating how employee brand identification or their engagement in brand-building behaviors might be shaped by service recovery processes could provide novel and practically relevant insights. Furthermore, exploring concepts such as employee grit or resilience, which refer to their ability to persevere and maintain a positive attitude in the face of challenging situations, could deepen our understanding of employee responses during service recovery. Examining the role of job meaningfulness, or the extent to which employees find their work purposeful and fulfilling, could also contribute to our understanding of how employees' perceptions and attitudes influence the recovery process. Finally, investigations of service employees' ethical dispositions and ethical behaviors in the context of SFR could offer insights into the potential impact of ethical considerations on service recovery outcomes. Studies exploring how ethical (unethical) behavior might enhance (undermine) the effectiveness of service recovery strategies would be of theoretical and practical relevance. By delving into these additional dimensions of service employee reactions, future research can expand our understanding of the complexities involved in service recovery.

Apology → Work Engagement? Figure 2 reveals a disconnection between the "work engagement" and "apology" keywords. Service employees play a crucial role in implementing service recovery strategies, and their level of work engagement can significantly impact the effectiveness of the recovery process. Therefore, it is worth exploring how offering apologies affects the work engagement of service employees. One possible outcome is that giving apologies may lead to feelings of shame or embarrassment among employees, diminishing their perceived job meaningfulness and undermining their overall work engagement. At the same time, offering apologies might also improve the relationship between employees and customers,

generating positive customer sentiments, which, in turn, could enhance service employees' work engagement. Thus, future research investigating the impact of different types of apologies, such as sincere apologies versus forced apologies, on work engagement would be valuable.

Double-Edge Swords? One potentially fruitful avenue for future SFR research is to simultaneously examine the effects of service recovery strategies on both customer and employee reactions. It is plausible that certain recovery strategies may generate "double-edged sword" effects, meaning they may positively impact one stakeholder group but negatively impact the other. For example, following apologies, customers might be more inclined to forgive, but employees may have to invest in more emotional labor, potentially leading to emotional exhaustion or work disengagement. Therefore, future studies could explore the potential double-edged sword effects of service recovery strategies, where positive customer reactions may coincide with negative employee reactions, or vice versa. By understanding these dynamics, organizations can develop thoughtful strategies that strike a balance, ensuring positive outcomes for both customers and employees in the service recovery process.

Research Opportunities Related to the Relationship Between Customer and Employee Reactions

SFR processes involve interactions between customers and frontline employees, with significant implications for firm-level outcomes. The literature on emotional contagion suggests that both customers and employees influence each other's affect, cognition, and behavior (Barsade 2002; Herrando and Constantinides 2021). For instance, a study by Jerger and Wirtz (2017) demonstrated that angry customers at fast food restaurants can immediately influence the emotional responses of employees, leading to expressed anger. Despite the significance of reciprocal relationships between customer reactions and employee reactions, our analysis, as depicted in Figure 3, indicates that this crucial conceptual linkage has received limited attention in the SFR literature to date.

Customer Anger ↔ Work Engagement? Our SNA findings, visually presented in Figure 2, reveals a notable gap between the "customer anger" and "work engagement" keywords. Future research may leverage emotional contagion theory (Barsade 2002; Herrando and Constantinides 2021) to explore the intricate, reciprocal relationship between the two constructs. It is conceivable that customer anger could strengthen or undermine employee work engagement. The strengthening effect occurs when employees are (intrinsically or extrinsically) motivated to address and resolve customer anger, using it as a driving force to improve their performance. Conversely, the undermining effect arises when employees perceive the anger as unfair, especially if they were not directly responsible for the service failure (Groth and Grandey 2012; Rupp and Spencer 2006). This perception of unfairness can dampen their work engagement, in the short-run and possibly the long-run.

On the other hand, service employee work engagement can also influence customer anger in various ways. The way employees approach customer issues (e.g., with patience and empathy, with a focus on protecting the firm's reputation) can mitigate or exacerbate customer anger. When employees demonstrate high levels of work engagement, they are more likely to address customer concerns with care and understanding, potentially reducing the intensity of customer anger. However, in some cases, employees may become overly protective of the organization's reputation and be confrontational, which can inadvertently fuel customer anger. As such, we believe that future research that helps unpack the complex and dynamic relationship between customer anger and employee work engagement would be valuable.

Exploring Contingencies (e.g., *Leadership, Online vs. Offline*). Researchers with an interest in this area can utilize Figure 2 as a tool to identify potential moderators that influence the relationship between customer and employee reactions. Our SNA results reveal an absence of connections between the keywords "leadership" and both "work engagement" and "customer anger." However, because different leadership styles, such as empowering leadership (Cheong et al. 2019) or directive versus participative leadership (House 1996), could have a notable impact, it is worth exploring the role of leadership style in this context. For example, certain leadership styles may mitigate potentially harmful effects of customer anger on employee work engagement. At the same time, other leadership styles may mitigate the potentially harmful effects of employee work disengagement on customer anger.

Furthermore, Figure 2 indicates a lack of connection between "online complaints" and both "work engagement" and "anger," suggesting a potential research gap. One possible argument is that the relationship between customer anger and employee work engagement may be weaker in an online setting compared to an offline setting. This could be due to employees being more "distant" and unable to observe nonverbal cues from customers, such as facial expressions and body language, and thus, being less influenced by them emotionally. However, an alternative perspective is that, compared to an offline setting, customer anger in an online setting could be more severe (e.g., more personal attacks given relative anonymity) and more persistent. Moreover, emotional contagion literature suggests that this anger can quickly spread on a larger scale (Herrando and Constantinides 2021), potentially leading to a more negative impact on employee work engagement.

Research Opportunities Related to the Impact of Service Recovery Strategy on Firm Outcomes

Firms embark on service recovery strategies with the goal of improving firm performance. Figure 3 shows four paths for this effect:

- (1) Service recovery strategy → customer reactions → Firm outcomes

- (2) Service recovery strategy → employee reactions → customer reactions → Firm outcomes
- (3) Service recovery strategy → employee reactions → Firm outcomes
- (4) Service recovery strategy → customer reactions → employee reactions → Firm outcomes

Of the paths above, the first path is the most commonly researched perspective in extant SFR literature, according to Figure 3. The other three paths are relatively understudied. Future research could utilize structural holes in Figure 2 to further explore these understudied paths. Below is just one example.

Double Deviation → Customer Reactions → Employee Reactions → Brand Equity. The structural hole between the keywords “double deviation” and “brand equity” highlights an area that has not been fully explored in existing SFR research. Double deviation refers to a situation wherein customer dissatisfaction intensifies following both a service failure and a failed attempt at service recovery (Bitner, Booms, and Tetreault 1990). Understanding the connection between double deviation and brand equity is crucial, as previous studies have shown that double deviation can trigger customer frustration and customer revenge (Surachartkumtonkun, McColl-Kennedy, and Patterson 2015; Ward and Ostrom 2016). If customer frustration and revenge are directed towards service employees, they can lead to employee frustration and burnout. These customer reactions and subsequent employee reactions may collectively dampen a company’s brand equity. Indeed, customers (e.g., Xie et al. 2019) and frontline employees (e.g., Badrinarayanan and Sierra 2018) are primary types of brand advocates for firms. Thus, exploring the mechanisms (including possible serial mediation) underlying the relationship between double deviation and brand equity can provide valuable insights into the consequences of failed service recoveries on the overall reputation and standing of a brand.

When SFR meets Wall Street. Another important variable that affects companies, lenders, and investors is firm value. Our

SNA results have revealed an interesting shift in the prominence of the online complaint/social media keyword. The transparency of these complaints on social media platforms is a notable distinction compared to offline complaints (Colm et al., 2017). As a result, these complaints become visible not only to consumers but also to investors and analysts, which can make investors more sensitive to service failures and service recovery strategies. Grégoire and Mattila (2021) have emphasized the need for future SFR studies to incorporate more quantitative analysis using rich data. Similarly, we propose that SFR researchers could effectively employ quantitative models, such as event studies and vector autoregression, to investigate how firm value is influenced by the implementation of service recovery strategies in response to these online and offline complaints.

Supplementary Analyses

Supplementary Analysis 1: Research Methods Used in SFR Research

In addition to research topics, the methodology employed in a study holds indispensable significance. To ensure a comprehensive analysis, we reviewed research methods used in SFR literature published between 1990 and 2022. Figure 5 serves as a visual representation, showcasing the distribution of research methods across four distinct periods.

Figure 5 unveils a prevailing trend in which most previous studies relied on a single research method, either empirical (e.g., survey, secondary data) or experimental. To overcome the limitations inherent in relying solely on a single research method (e.g., limited external validity of experimental research, limited internal validity of survey research), several recent SFR articles have adopted a multi-faceted approach, utilizing both empirical and experimental research methods to augment the validity of their findings (e.g., Casidy et al. 2021; Hall and Hyodo 2022; Herhausen et al. 2023). Figure 5 illuminates an upward trajectory in the percentage of articles employing multiple research methods during the 2020-2022 period. Given the inherent nature of SFR as a dynamic process (e.g., Zhu and Zolkiewski 2015), relying solely on cross-sectional data may

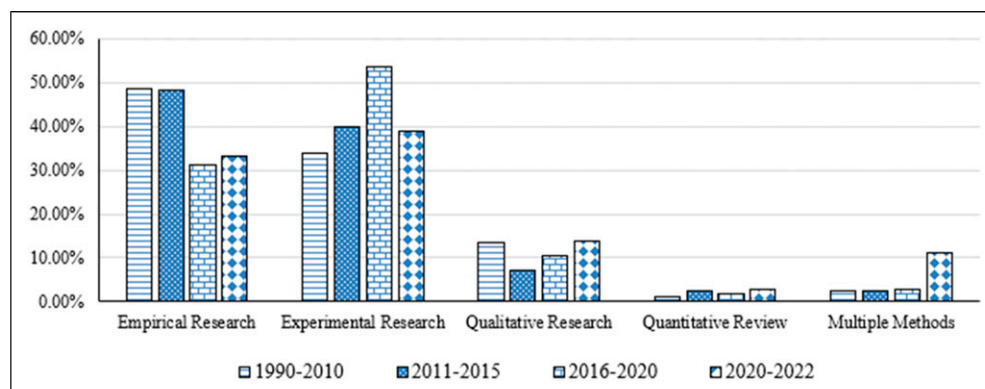


Figure 5. Distribution of research methods in SFR literature from 1990 to 2022. Note: qualitative research includes case studies, integrative reviews, and interviews; quantitative review includes meta-analysis, topic modeling, and bibliometric analysis.

restrict researchers' ability to capture the true progression of SFR events over time. In order to shed light on the process-oriented aspect of SFR, select studies have employed longitudinal data. For instance, [Kanuri and Andrews \(2019\)](#) analyzed data spanning 39 months from a prominent U.S. newspaper firm, while [Grégoire et al., 2018](#) conducted longitudinal field studies and experiments to investigate the phenomenon of customer revenge.

Supplementary Analysis 2: Top Keywords in the 2020–2022 SFR Literature

To gain further insights into the latest research trends, we conducted an analysis of the SFR literature from 2020 to 2022, with a focus on identifying the prominent keywords employed during this period. We found a significant overlap between the top keywords in this period and those from the 2016–2020 period. Keywords such as “online complaint” and “negative word-of-mouth” continued to dominate the discourse in the field during 2020–2022. However, our analysis revealed the emergence of three new keywords not as prevalent in the previous period. These new keywords include “failure scenario” (e.g., group versus individual scenarios, [Albrecht et al. 2019](#)), “process failure” ([Borah, Prakhya, and Sharma 2020](#)), and “self-service technology” ([Sheehan, Jin, and Gottlieb 2020](#)).

Scholars began to pay attention to SFR issues concerning self-service technology in the early 2000s ([Meuter et al. 2000](#); [Bitner, Ostrom, and Meuter 2002](#)). However, it was not until the 2020–2022 period that the keyword “self-service technology” emerged as a prominent term in our list of top keywords. We attribute this shift to the rising implementation of AI-enabled self-services and service robots by companies ([Huang and Philp 2021](#); [Jörling, Böhm, and Paluch 2019](#)). This development opens up a promising avenue for research in the field. Recent studies have begun to delve into consumer responses to service failures induced by AI (Artificial Intelligence) (e.g., perceived responsibility, propensity to share negative word-of-mouth, [Huang and Philp 2021](#); [Jörling, Böhm, and Paluch 2019](#)). Looking ahead, future research holds the potential to shed light on several key issues: (1) examining both consumer and employee responses to service failures induced by AI, (2) investigating the underlying drivers behind these AI-induced service failures, (3) identifying effective recovery strategies tailored specifically to AI-induced service failures, and (4) exploring the utilization of AI itself in the process of recovering from AI-induced service failures.

Supplementary Analysis 3: Citation Analysis of the Most Recent SFR Literature

While our examination of structural holes has shed light on emerging research topics, generating specific research directions to propel the field forward is no simple task. To ascertain the emergence of novel research topics and to complement our co-word analysis, we embarked upon a citation analysis of the

SFR literature from 2020 to 2022, specifically targeting highly cited articles published after 2017. This analysis yielded 25 articles of significant impact (refer to [Table 1WA in the blue appendix](#)). After examining these 25 articles, we identified the following pioneering research areas:

SFR in the Online/Social Media Landscape. This is consistent with our finding #1 in co-word analysis. Given the increasing prevalence of services being conducted through online interfaces, particularly during the global COVID-19 pandemic, there has been a notable shift towards exploring SFR within the digital domain (e.g., [Herhausen et al. 2023](#); [Hogreve, Bilstein, and Hoerner 2019](#)).

Specific Recovery Strategies in Diverse Contexts. Scholars increasingly acknowledge that a one-size-fits-all recovery strategy does not exist. Recent investigations have delved into the interplay between various recovery strategies (e.g., [Borah, Prakhya, and Sharma 2020](#)), the examination of different variations of a particular recovery strategy, such as group versus individual compensation (e.g., [Albrecht et al. 2019](#)), and the effectiveness of recovery strategies within different contextual settings (e.g., [Borah, Prakhya, and Sharma 2020](#); [Crisafulli and Singh 2017](#); [Hogreve, Bilstein, and Mandl 2017](#)).

Co-Creation of Recovery Strategies. The SFR literature has actively advocated for customer participation in the recovery process ([Van Vaerenbergh, Hazée, and Costers 2018](#)). However, an area that remains inadequately explored pertains to how co-creation recovery strategies can be effectively implemented in practice.

Positive Customer Emotions. Since 2010, customer emotions have garnered significant attention within the field. However, the subsequent decade primarily focused on examining negative customer emotions such as anger, disappointment, and frustration. More recently, there has been a notable surge in exploring positive customer emotions, with one prominent area of investigation revolving around customer forgiveness ([Harrison-Walker 2019](#); [Tsarenko, Strizhakova, and Otnes 2019](#)).

Conclusions

We conducted a co-word analysis of SFR literature spanning three decades. Our aim was to enhance understanding of the literature and augment existing bibliometric reviews by exploring the specific concepts and variables examined in SFR research. Through co-word analysis and SNA, we uncovered nuanced insights that contribute to a deeper understanding of SFR. Additionally, taking a longitudinal perspective allowed us to trace the evolution of SFR literature and identify changes in research emphases and emerging topics. For example, our analysis reveals a reduced emphasis on service employee-related topics in the more recent period of 2016–2020 compared to the earlier period of 1990–2009. This indicates a shift in research interests over time. Furthermore, our longitudinal

analysis revealed the prevalence of specific recovery strategies, customer reactions, and employee reactions during each time period. For instance, in the most recent period, there was a notable increase in research attention paid to positive customer reactions, including forgiveness and positive emotion. By “zooming in” and offering a more fine-grained and longitudinal perspective, our study complements previous bibliometric reviews that summarize the foundational literature’s overarching themes.

Building on the idea that combining different forecasting methods and different information sources can improve forecasting accuracy (Armstrong 2001), our use of SNA to derive future directions also complements existing bibliometric reviews of the SFR literature. Existing reviews propose future directions based on the authors’ knowledge about the literature and their trend predictions. In a data-driven manner, we complement this approach to future research by (objectively) deriving specific future research opportunities and questions based on structural holes in SFR keyword networks. Thus, we can propose specific research questions that address observed gaps within the SFR literature. These questions focus on exploring understudied direct effects and their moderators and mediators within the interconnected domains of service recovery strategy, customer reactions, employee reactions, and firm performance.

In addition to our main study, we conducted several supplementary analyses to deepen our understanding of SFR research. These supplementary analyses focused on exploring the research methods utilized in SFR studies and identifying recently trending topics in SFR research published between 2020 and 2022. By integrating insights from existing bibliometric reviews that employed co-citation analysis and combining them with our main analysis involving co-word analysis, SNA, and the three additional analyses, we augment understanding of the extant SFR literature and its future trajectory.

Limitations

While this research provides valuable insights, it is important to acknowledge certain limitations. First, our data collection might not have captured every study on the topic of SFR, as our search was limited to English-language articles in specific journals and databases. Future research could include articles published in other languages to ensure a more comprehensive coverage. Second, our focus on high-frequency keywords in the co-word analysis might have overlooked important but less frequent keywords, potentially missing out on emerging concepts or ideas in the field. Additionally, due to space constraints, we were unable to discuss every keyword and its changes in the co-word analysis results. Researchers interested in this area could explore the additional insights presented in Table 2, such as the declining popularity of the “critical incident (method)” keyword over time.

Furthermore, the future research opportunities proposed in this study primarily stem from the co-word analysis of the 2016–2020 SFR literature and the supplementary analyses of the 2020–2022 literature. However, researchers should not

overlook the potential research opportunities illuminated by older SFR literature. An interesting observation from the keyword network of the 2010–2015 period is the lack of connection between emotional intelligence and intangible compensation. This prompts the question of whether service employees with higher (lower) emotional intelligence might be more (less) motivated by intangible rewards. This example, we believe, highlights the potential of keyword networks—even in older time periods—to identify possibly fruitful research opportunities for the future.

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Supplemental Material

Supplemental material for this article is available online.

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